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Stefano Martiniani

Academic Positions

Assistant Professor	**New York University** Dept. of Physics, Center for Soft Matter Research Dept. of Chemistry, Simons Center for Computational Physical Chemistry Courant Institute of Mathematical Sciences Center for Neural Science Center for Data Science (Affiliated)	2022 – present
Assistant Professor	**University of Minnesota - Twin Cities** Dept. of Chemical Engineering and Materials Science Dept. of Physics (Affiliated) Data Science (Affiliated)	2019 – 2021

Education

BSc Chemistry with First Class Honors	**Imperial College London**	2012
MPhil Scientific Computing with Distinction	**University of Cambridge**	2013
PhD Chemistry Outstanding Thesis Prize	**University of Cambridge**	2017
Postdoctoral – Physics	**New York University**	2017 – 2019

Academic summary

Selected Awards	AFOSR YIP, NSF CAREER, IUPAP Interdisciplinary Early Career Scientist Prize, CZI Neuroscience Pairs Pilot Award, Simons Foundation Faculty Fellowship, Outstanding PhD Thesis Prize
Publications	41 research articles (8 lead author, 25 corresponding author, 20 led by trainees), 6 workshop papers (3 spotlight), 5 extended abstracts, 1 commentary, 1 editorial, 2 provisional US patents.
Media	Research highlighted on PRL cover, PCCP cover, Nature, Nature Materials, Physics (1, 2, 3), Physics Today (1, 2), New Scientist, Science & Vie, Sky News, ANSA.
Presentations	79 past talks (59 invited), 11 posters.
Funding	AFOSR FA9550-25-1-0359 (PI), NSF DMR 2443027 (PI), NSF OAC 2311632 (PI), NIH R01 MH137669 (PI), NIH R01 EY035242 (PI), CZI 2024-338565 (PI), NSF III 2132995 (PI), NSF CESER 2039575 (Co-PI), NIH R01 GM146372 (Co-I), CZI NS-CS-39 (Co-I). Total PI/co-PI extramural funding: $12,803,189 ($11,573,999 as PI).
Leadership	Lead PI of FERMat project, involving 8 investigators across 4 universities (NYU, UMN, UF, BYU) and 1 industry partner (AWS); co-PI of ColabFit Exchange; Program Chair AI4Mat Workshop (NeurIPS x2, ICLR x2);

	Organizer NYC AI4Chemistry Summit (2025, 2026); Co-founder and member of organizing committee KIMReview;.
Teaching Innovation	Developed UMN's first non-CS/Stats ML course "Machine Learning for Chemical Sciences" and graduate course "Physics of Neural Systems" at NYU.
Mentorship	Currently leading a group of 8 graduate students, 9 postdocs/research scientists, 1 software developer, 1 MS, 1 UG. **Past trainees:** 2 PhDs in ChEn, 3 PhDs in Mathematics, 1 PhD Physics, 2 Postdoc, 1 master, 15 undergraduates (3 published), 4 high-school students (1 published), 3 ColabFit interns (2 published).

Honors & Awards

Entropy Young Investigator Award	MDPI *Entropy*	2025
AFOSR YIP Award	AFOSR/RTB	2025
NSF CAREER Award	NSF DMR-CMMT	2024
The David Iakobachvili Interdisciplinary Science Research Award	New York University	2024
Gallery of Soft Matter Prize	APS Division of Soft Matter	2024
Neuroscience Pairs Pilot Project Award	Chan Zuckerberg Initiative	2024
Interdisciplinary Early Career Scientist Prize	Int. Union of Pure and Applied Physics	2023
Simons Foundation Faculty Fellowship	New York University	2022 – 2025
Outstanding Thesis Prize	Dept. of Chemistry, U. of Cambridge	2017
Gates Cambridge Scholarship	University of Cambridge	2013 – 2017
Benefactors Scholarship	St. John's College, U. of Cambridge	2013 – 2017
Prize for Best Physical Chemistry Research Project (BS/MS Thesis)	Dept. of Chemistry, Imperial College	2012
Undergraduate Research Fellowship	Imperial College London	2012

Extramural Funding

1. **Role:** PI; **PD/PI:** Martiniani; **Title:** "Advancing Light Manipulation and Optical Computing with Engineered Disordered Photonic Metamaterials"; **Program:** AFOSR YIP; **Funds:** $449,420; **Duration:** (Sept 2025 - Sept 2028) **Award number:** FA9550-25-1-0359.
2. **Role:** PI; **PD/PI:** Martiniani; **Title:** "CAREER: Quantifying the Complexity of Materials Landscapes by Basin Sampling"; **Program:** NSF DMR-CMMT; **Funds:** $700,000; **Duration:** 5 years (Mar 2025 - Feb 2030); **Award number:** 2443027.
3. **Role:** PI; **PD/PI:** Heeger (contact), Martiniani; **Title:** "Oscillatory Recurrent Gated Neural Integrator Circuits (ORGaNICs): a unified framework for neural dynamics and human cognition"; **Program:** NIH NIMH R01 (multi-PI); **Funds:** $3,670,550; **Duration:** 5 years (Jul 2024 - Jun 2029); **Award number:** R01MH137669.

4. **Role:** PI; **PD/PI:** Martiniani; **Title:** "Theme 2: AI Institute for Extreme Computing Materials (AI-XCoM)"; **Program:** NYU Mega Grants Initiative Seed Fund; **Funds:** $25,000; **Duration:** 11 months (Feb 2024 - Dec 2024);
5. **Role:** PI; **PD/PI:** Martiniani (contact), Fenton; **Title:** "Machine Learning the Biomolecular Basis of Memory Persistence"; **Program:** CZI Neuroscience Pairs Pilot Project Awards; **Funds:** $200,000; **Duration:** 2 years (March 2024 - March 2025); **Award number:** 2024-338565.
6. **Role:** PI; **PD/PI:** Martiniani; **Title:** "GOALI: Frameworks: At-Scale Heterogeneous Data based Adaptive Development Platform for Machine-Learning Models for Material and Chemical Discovery"; **Program:** NSF OAC; **Funds:** $4,500,000; **Duration:** 5 years (Oct 2023 - Sept 2028); **Award number:** 2311632.
7. **Role:** PI; **PD/PI:** Heeger (contact), Martiniani; **Title:** "Recurrent circuit model of neural response dynamics in V1"; **Program:** NIH NEI R01 (multi-PI); **Funds:** $1,879,804; **Duration:** 4 years (Sept 2023 - Sept 2027); **Award number:** R01EY035242.
8. **Role:** PI; **PD/PI:** Martiniani; **Title:** "EAGER: Quantifying the error landscape of deep neural networks"; **Program:** NSF III; **Funds:** $149,225 ; **Duration:** 2 years (Oct 2021 - Aug 2024); **Award number:** 2132995.
9. **Role:** co-PI; **PD/PI:** Tadmor; **Title:** "Data CI Pilot: CI-Based Collaborative Development of Data-Driven Interatomic Potentials for Predictive Molecular Simulations"; **Program:** NSF CESER; **Funds:** $1,127,993; **Duration:** 2 years (Oct 2020 - Sept 2024); **Award number:** 2039575.
10. **Role:** co-I; **PD/PI:** Kumar, Fenton; **Title:** "PKMzeta organization in persistent memory and non-memory identified spines"; **Program:** CZI Neuroscience Collaboration Supplements; **Funds:** $30,000; **Duration:** 1 year (Dec 2024 - Dec 2025); **Award number:** NS-CS-39.
11. **Role:** co-I; **PD/PI:** Hackel; **Title:** "Engineering Protein Developability"; **Program:** NIH GM R01; **Funds:** subaward $71,197 (total $1,815,456); **Duration:** 4 years (Oct 2022-2026); **Award number**: GM146372.

Publications [*trainee, †contributed equally, (co-)corresponding author]

Preprints/submitted

41. L.V. Luzzatto, M. Casiulis*, **S. Martiniani**, I. Kovacs, "Spatial and Temporal Cluster Tomography of Active Matter", *arXiv preprint* arXiv:2511.09444 (2025)
40. P. Prakash, J. B. Gibson, Z. Li, G. Di Gianluca, J. Esquivel, E. Fuemmeler*, B. Geisler, J. S. Kim, A. Roitberg, E. B. Tadmor, M. Liu, **S. Martiniani**, G. R. Stewart, J. J. Hamlin, P. J. Hirschfeld, R. G. Hennig, "Guided Diffusion for the Discovery of New Superconductors", *arXiv preprint* arXiv:2509.25186 (2025)
39. F. Barrows, G. Zhang*, S. Anand*, Z. Chen*, J. Lin, A. Desai, **S. Martiniani**, F. Caravelli, "A unifying approach to self-organizing systems interacting via conservation laws", *arXiv preprint* arXiv:2507.02575 (2025)

38. A. Pal*, S. Rawat*, D.J. Heeger, **S. Martiniani**, "Hierarchical Neural Circuit Theory of Normalization and Inter-areal Communication", bioRxiv 2025.07.15.664935 (2025)

37. F. Morone*, S. Rawat*, D.J. Heeger, **S. Martiniani**, "Stabilization of recurrent neural networks through divisive normalization", bioRxiv 2025.05.16.654567 (2025)

36. J. Han*, A. Grau-Perales, R.M. Harris, H. Kao, A. Pal*, J.M. Alarcon, T. Sacktor, **S. Martiniani**, H.A. Hofmann, A.A. Fenton, "Persistently increased expression of PKMς and unbiased gene expression profiles identify hippocampal molecular traces of a long-term active place avoidance memory and 'shadow' proteins", bioRxiv 2025.04.28.651143 (2025)

35. G. Zhang*, **S. Martiniani**, "Absorbing state dynamics of stochastic gradient descent", *arXiv preprint* arxiv:2411.11834 (2024)

34. P. Suryadevara*, M. Casiulis*, **S. Martiniani**, "The Basins of Attraction of Soft Sphere Packings are Not Fractal", *arXiv preprint* arXiv:2409.12113 (2024)

33. T.V. Phan, S. Li, D. Ferreris, R. Morris, J. Bos, B. Guo, **S. Martiniani**, P. Chaikin, Y.G. Kevrekidis, R.H. Austin, "Social Physics of Bacteria: Avoidance of an Information Black Hole", *arXiv preprint* arXiv:2401.16691 (2024)

Peer-reviewed articles

32. S. Anand*, G. Zhang*, **S. Martiniani**, "Emergent universal long-range structure in random-organizing systems", *Nat. Comm.*, accepted (*arXiv preprint* arXiv:2505.22933) (2026)

31. C Zeng, J Jin, G Karypis, M Transtrum, EB Tadmor, RG Hennig, A. Roitberg, **S. Martiniani**, M. Liu, "PropMolFlow: Property-guided Molecule Generation with Geometry-Complete Flow Matching", *Nat. Comp. Sci.,* accepted (*arXiv preprint* arXiv:2505.21469) (2026)

30. G. Zhang*, D.J. Heeger*, **S. Martiniani**, "Contrastive Self-Supervised Learning As Neural Manifold Packing", NeurIPS 2025, accepted (arXiv:2506.13717) (2025)

29. M. Martirossyan*, T. Egg*, P. Höllmer*, G. Karypis, M. Transtrum, A. Roitberg, M. Liu, R. Hennig, E.B. Tadmor, **S. Martiniani**, "All that structure matches does not glitter", NeurIPS 2025, accepted (*arXiv preprint* arXiv:2509.12178) (2025)

28. M. Casiulis*, A. Shih*, **S. Martiniani**, "Gyromorphs: a new class of functional disordered materials", *Phys. Rev. Lett.*, 135, 196101 (2025)

27. M. Casiulis*, E. Arbel, Y. Lahini, **S. Martiniani**, N. Oppenheimer, M. Yah Ben Zion, "A geometric condition for robot-swarm cohesion and cluster-flock transition", *Proc. Natl. Acad. Sci.,* 122 (37) e2502211122 (2025)

26. P. Hoellmer*, T. Egg*, M.M. Martirossyan*, E. Fuemmeler*, Z. Shui, A. Gupta*, P. Prakash, A. Roitberg, M. Liu, G. Karypis, M. Transtrum, R.G. Hennig, E.B. Tadmor, **S. Martiniani**, "Open Materials Generation with Stochastic Interpolants", *Proc. 42nd Int. Conf. Mach. Learn. (ICML), PMLR* 267 (2025)

25. Y. Wang, K. Takaba, M.S. Chen, M. Wieder, Y. Xu, T. Zhu, J.Z.H. Zhang, A. Nagle, K. Yu, X. Wang, D.J. Cole, J.A. Rackers, K. Cho, J.G. Greener, P. Eastman, **S. Martiniani**, M.E. Tuckerman, "On the

design space between molecular mechanics and machine learning force fields", *Appl. Phys. Rev.* 12, 021304 (2025)

24. R. Jacobs, D. Morgan, S. Attarian, J. Meng, C. Shen, Z. Wu, C. Xie, J.H. Yang, N. Artrith, B. Blaiszik, G. Ceder, K. Choudhary, G. Csanyi, B. Deng, R. Drautz, J. Godwin, V. Honavar, O. Isayev, A. Johansson, **S. Martiniani**, S.P. Ong, I. Poltavsky, K.J. Schmidt, S. Takamoto, A. Thompson, J. Westermayr, "A Practical Guide to Machine Learning Interatomic Potentials–Status and Future", *Curr. Opin. Solid State Mater. Sci.*, 35, 101214 (2025)

23. S Anand*, X. Ma, S. Guo, **S. Martiniani**†, X. Cheng†, "Transport and Energetics of Bacterial Rectification", *Proc. Natl. Acad. Sci.*, 121(52), e2411608121 (2024)

22. S. Rawat*, D. Heeger, **S. Martiniani**, "Unconditional stability of a recurrent neural circuit implementing divisive normalization", *Adv. Neural. Inf. Process. Syst. 38* (NeurIPS 2024), (arXiv:2409.18946) (2024)

21. S. Rawat*, **S. Martiniani**, "Element-wise and Recursive Solutions for the Power Spectral Density of Biological Stochastic Dynamical Systems at Fixed Points", *Phys. Rev. Res.*, 6, 043179 (2024)

20. A. Shih*, M. Casiulis*, **S. Martiniani**, "Fast Generation of Spectrally-Shaped Disorder", *Phys. Rev. E* 110, 034122 (2024)
    - Highlighted as **Editor's Suggestion** in *Phys. Rev. E*
    - Awarded **Gallery of Soft Matter Prize** and highlighted in *Physics,* 17, 41 (2024).
    - Highlighted in **"Old Movie Demos New Tech"**, *Physics,* 17, 134 (2024).

19. J.A. Vita*, E.G. Fuemmeler*, A. Gupta*, G.P. Wolfe*, A.Q. Tao*, R.S. Elliott, **S. Martiniani**, E. B. Tadmor, "ColabFit Exchange: open-access datasets for data-driven interatomic potentials", J. Chem. Phys. 159, 154802 (2023)

18. A.W. Golinski*, Z.D. Schmitz*, G.H. Nielsen, B. Johnson*, D. Saha*, S. Appiah*, B.J. Hackel, **S. Martiniani**, "Predicting and Interpreting Protein Developability via Transfer of Convolutional Sequence Representation", *ACS Synth. Biol.* 12, 9, 2600 (2023)

17. M. Casiulis*, **S. Martiniani**, "When you can't count sample! Computable entropies beyond equilibrium from basin volumes", *Papers in Physics,* 15, 150001 (2023).
    - Invited perspective in "Focus Series on Challenges in Granular Matter", Editors: K. Daniels, L. Pugnaloni, J. Zhao.

16. C. Anzivino, M. Casiulis*, T. Zhang*, A.S. Moussa, **S. Martiniani**, A. Zaccone, "Estimating random close packing in polydisperse and bidisperse hard spheres via an equilibrium model of crowding", *J. Chem. Phys.,* 158, 044901 (2023)

15. S. Ro, B. Guo, A. Shih*, T.V. Phan, R.H. Austin, D. Levine, P.M. Chaikin, **S. Martiniani**, "Model-Free Measurement of Local Entropy Production and Extractable Work in Active Matter", *Phys. Rev. Lett.* 129, 220601 (2022).
    - **Cover article** for *Phys. Rev. Lett.*
    - Highlighted as **Editor's Suggestion** in *Phys. Rev. Lett.*
    - Highlighted in **"Measuring Entropy in Active-Matter Systems"**, *Physics* 15, 179 (2022)

14. A. W. Golinski*, K.M. Mischler, S. Laxminarayan, N. Neurock, M. Fossing, H. Pichman, **S. Martiniani**, B.J. Hackel, "High-Throughput Developability Assays Enable Library-Scale Identification of Producible Protein Scaffold Variants", *Proc. Natl. Acad. Sci.*, 118, 23 (2021)

13. A. Cavagna, P.M. Chaikin, D. Levine, **S. Martiniani**, A. Puglisi, M. Viale, "Vicsek Model by Time-Interlaced Compression: a Dynamical Computable Information Density", *Phys. Rev. E* 103, 062141 (2021)

12. **S. Martiniani**, Y. Lemberg, P.M. Chaikin, D. Levine, "Correlation lengths in the language of computable information", *Phys. Rev. Lett.*, 125, 170601 (2020)

11. **S. Martiniani**, P.M. Chaikin, D. Levine, "Quantifying hidden order out of equilibrium", *Phys. Rev. X,* 9, 011031 (2019)
    - Highlighted in "**File compression uncovers hidden order**", *Physics Today* (2019).

10. **S. Martiniani**, K.J. Schrenk, K. Ramola, B. Chakraborty, D. Frenkel, "Numerical test of the Edwards conjecture shows that all packings become equally probable at jamming", *Nature Physics*, 13, 848–851 (2017)
    - Highlighted in "**Intuition harnessed in the name of particle packing**", *Nature*, 546, 575 (2017).
    - Highlighted in "**A thermodynamic theory of granular materials endures**", *Physics Today* 70, 9 (2017).
    - Highlighted in "**Material Witness: A jammy guess?**", *Nature Materials*, 15, 1227 (2016).

9. D. Frenkel, K.J. Schrenk, **S. Martiniani** "Monte Carlo sampling for stochastic weight functions", *Proc. Natl. Acad. Sci.,* 114, 27 (2017)

8. A.J. Ballard, R. Das, **S. Martiniani**, D. Mehta, L. Sagun, J.D. Stevenson, D.J. Wales, "Energy Landscapes for Machine Learning", *Phys. Chem. Chem. Phys.,* 19, 12585 (2017)
   - **Cover article** for *Phys. Chem. Chem. Phys.*

7. **S. Martiniani**, K.J. Schrenk, J.D. Stevenson, D. J. Wales, D. Frenkel, "Structural analysis of high dimensional basins of attraction", *Phys. Rev. E* 94, 031301 (2016)

6. **S. Martiniani**, K.J. Schrenk, J.D. Stevenson, D.J. Wales, D. Frenkel, "Turning intractable counting into sampling: computing the configurational entropy of three-dimensional jammed packings", *Phys. Rev. E* 93, 012906 (2016)

5. A.J. Ballard, **S. Martiniani**, J.D. Stevenson, S. Somani, D.J. Wales, "Exploiting the potential energy landscape to sample free energy", *WIREs Comput. Mol. Sci.* 5, 273 (2015)

4. **S. Martiniani**, J.D. Stevenson, D.J. Wales, D. Frenkel, "Superposition Enhanced Nested Sampling", *Phys. Rev. X* 4, 031034 (2014)

3. C. Magistris†, **S. Martiniani**†, N. Barbero, J. Park, C. Benzi, A. Anderson, C.H. Law, C. Barolo, B.C. O'Regan, "Near-infrared absorbing squaraine dye with extended π conjugation for dye-sensitized solar cells", *Renewable Energy* 60, 672 (2013)

2. C.E. Richards, A.Y. Anderson, **S. Martiniani**, C. Law, B.C. O'Regan, "The Mechanism of Iodine Reduction by TiO2 Electrons and the Kinetics of Recombination in Dye-Sensitized Solar Cells", *J. Phys. Chem. Lett.* **3**, 1980 (2012)

1. **S. Martiniani**, A.Y. Anderson, C. Law, B.C. O'Regan and C. Barolo, "New insight into the regeneration kinetics of dye sensitized solar cells", *Chem. Commun.* 48, 2406 (2012)

Peer-reviewed workshop papers

6. P. Prakash, J. B. Gibson, Z. Li, G. Di Gianluca, J. Esquivel, E. Fuemmeler, B. Geisler, A. Roitberg, E. B. Tadmor, M. Liu, **S. Martiniani**, G. R. Stewart, J. Hamlin, P. Hirschfeld, R. Hennig, "Inverse Design of Novel Superconductors via Guided Diffusion", AI for Accelerated Materials Design - NeurIPS 2025, link
   - Selected as **Spotlight**

5. C. Zeng, J. Jin, P. Prakash, G. Karypis, M. Transtrum, E.B. Tadmor, R. Hennig, A. Roitberg, **S. Martiniani**, M. Liu, MolGuidance: A Comparative Study of Guidance Methods for Conditional Molecule Generation, Generative AI and Biology (GenBio) Workshop - ICML 2025, link

4. P. Hoellmer*, T. Egg*, M.M. Martirossyan*, E. Fuemmeler*, Z. Shui, A. Gupta*, P. Prakash, A. Roitberg, M. Liu, G. Karypis, M. Transtrum, R.G. Hennig, E.B. Tadmor, **S. Martiniani**, "Open Materials Generation with Stochastic Interpolants", AI for Accelerated Materials Design - ICLR 2025, link
   - Selected as **Spotlight**

3. E. Fuemmeler*, G. Wolfe*, A. Gupta*, J.A. Vita, E.B. Tadmor, **S. Martiniani**, "Advancing the ColabFit Exchange towards a Web-scale Data Source for Machine Learning Interatomic Potentials", AI for Accelerated Materials Design - NeurIPS 2024, link
   - Selected as **Spotlight**

2. C. Gonzales, E. Fuemmeler*, E.B. Tadmor, **S. Martiniani**, S. Miret , "Benchmarking of Universal Machine Learning Interatomic Potentials for Structural Relaxation", AI for Accelerated Materials Design - NeurIPS 2024, link

1. A. Gupta*, E.B. Tadmor, **S. Martiniani**, "KUSP: Python server for deploying ML interatomic potentials", AI for Accelerated Materials Design - Vienna 2024, link

Peer-reviewed extended abstracts

5. A. Pal*, S. Rawat*, D. Heeger, **S. Martiniani**, "Multi-stage Cortical Recurrent Circuit Implementing Normalization". CCN Abstracts (2024), link.

4. S. Rawat*, D. Heeger, **S. Martiniani**, "A comprehensive large-scale model of primary visual cortex (V1)". CCN Abstracts (2024), link.

3. G. Zhang*, **S. Martiniani**, "Neural manifold packing by stochastic gradient descent", CCN Abstracts (2024), link

2. S. Rawat*, D. Heeger, **S. Martiniani**, "A comprehensive large-scale model of primary visual cortex (V1)". Cosyne Abstracts (2024), link.

1. S. Rawat*, D. Heeger, **S. Martiniani**, "Coherence influences the dimensionality of communication subspaces". Cosyne Abstracts (2023), link.

Commentaries and Editorials

1. **S. Martiniani**, "Bit-propelled Active Matter". Journal Club for Condensed Matter Physics (2023), link.

Editorials

1. S. Miret, M. Skreta, G. Wellawatte, **S. Martiniani**, N. M. A. Krishnan, G. Karypis, K. M. Jablonka, "Perspective on Artificial Intelligence for Accelerated Materials Design (AI4Mat) Workshops in 2024", Mach. learn.: Sci. Technol., 6 (4), 040201 (2025)

Ph.D. thesis

S. Martiniani, "On the complexity of energy landscapes: algorithms and a direct test of the Edwards conjecture", University of Cambridge (2017)
- Awarded "Outstanding Thesis Prize", Dept. of Chemistry, University of Cambridge

Patents

1. A. Shih, M. Casiulis, **S. Martiniani**, "SYSTEM, METHOD AND COMPUTER-ACCESSIBLE MEDIUM FOR ACCELERATED GENERATION OF STATISTICALLY CORRELATED POINT STRUCTURES", Provisional Patent Application *63/651,613* (May 2024)
2. A. Shih, M. Casiulis, **S. Martiniani**, "SYSTEM, METHOD, AND COMPUTER-ACCESSIBLE MEDIUM FOR GENERATING FUNCTIONAL CORRELATED DISORDERED MATERIALS", Provisional Patent Application *63/811,349* (May 2025)

Talks (*invited)

Upcoming

1. * Seminar, Dept. of Chemical Engineering, Princeton University, Mar 2026
2. * Talk, CECAM Flagship Workshop: "Physics-aware machine learning for molecules and materials", Jun 2026
3. * Invited Lecture, Erice workshop on "Self-organized hyperuniformity in and out of equilibrium", Erice, Jul 2026

Past

1. * Talk, AFOSR Electromagnetic Portfolio Review, Jan 2026
   Title: "Gyromorphs: a new class of functional disordered materials"
2. * Invited Lecture, Lorentz Center workshop: AI for Secure, Safe, and Sustainable Molecules and Materials, Leiden, Nov 2025
   Title: "Daedalus' Design: The Art of Learning from Nature"
3. Talk, Workshop: "Canonical Neural Computations for Vision, Attention and Working Memory", Park City (UT), Nov 2025
   Title: "Stabilization of recurrent neural networks through divisive normalization"
4. * Invited Lecture, Reimagining the Semiconductor Future: From Atoms to Foundation Models, San Francisco, Oct 2025
   Title: *"Reimagining microelectronic materials discovery with generative AI"*

5. * Seminar, Vassar College, New York, Oct 2025
   Title: *"Though This Be Disorder, Yet There Is Order in't"*
6. * Mathematics in Physics Seminar, Dept. of Mathematics, MIT, Oct 2025
   Title: *"Though This Be Disorder, Yet There Is Order in't"*
7. * Seminar, Center for Computational Biology, Flatiron Institute, Sept 2025
   Title: "*Learning as Manifold Packing*"
8. * Invited Lecture, CAFE Workshop, AutoML Conference, NY, Sept 2025
   Title: *"Property-guided generative AI for molecules and materials"*
9. Talk, StatPhys29, Florence, July 2025
   Title: *"Emergent universal long-range structure in random-organizing systems"*
10. * Custom-Made Tech, Italian Trade Agency, New York, April 2025
    Title: *"Towards the AI Industrial Revolution"*
11. * AI in Science Seminar, University of Pennsylvania, Philadelphia, March 2025
    Title: *"Learning as Manifold Packing"*
12. * Physical Chemistry Seminar, Department of Chemistry, University of Florida, February 2025
    Title: *"Though This Be Disorder, Yet There Is Order in't"*
13. * Invited Lecture, 64th Sanibel Symposium, St. Augustine Beach (FL), February 2025
    Title: *"Open Materials Generation with Stochastic Interpolants"*
14. * Invited Lecture, Berkeley Statistical Mechanics Meeting, UC Berkeley, January 2025
    Title: *"Absorbing State Dynamics of Stochastic Gradient Descent"*
15. Seminar, Dept. of Chemistry, New York University, November 2024
    Title: *"Though This Be Disorder, Yet There Is Order in't"*
16. * Flash talk, The AI Alliance NYC meetup, Nov 2024
    Title: *"Generative AI for Materials Design"*
17. * IUPAP Interdisciplinary Early Career Scientist Prize, The 33rd IUPAP General Assembly, Haikou, China, October 2024 (remote)
    Title: *"Play. Pause. Rewind. Measuring local entropy production and extractable work in active matter"*
18. Computational Mathematics Seminar, Courant Institute of Mathematical Sciences, October 2024
    Title: *"The other side of entropy"*
19. * Swartz Seminar, Center for Neural Science, New York University, September 2024
    Title: *"Dynamic Normalization as a Theoretical Framework for Brains and Machines"*
20. * Energy Landscapes 2024, Lovran, Croatia, August 2024
    Title: *"Fast Generation of Spectrally Shaped Disorder"*
21. * ACS Fall Meeting, Symposium: Elevating Biomolecular Simulations With Data-Driven Approaches, Denver (CO), August 2024
    Title: *"Prediction and exploration of protein developability landscapes"*

22. * Soft, Living and Active Matter (SLAM) Seminar, Institute of Science and Technology Austria (ISTA), Vienna, July 2024
    Title: *"The other side of entropy"*
23. * Fireside Chat: AI Across Biological Scales, CZI Neuroscience Meeting, Monterey (CA), July 2024
    Title: *"AI Across Biological Scales: From Gene Expression to Behavior"*
24. * Chemical Engineering Seminar, California Institute of Technology, May 2024
    Title: *"The other side of entropy"*
25. * Physical Chemistry Seminar, University of California Irvine, May 2024
    Title: *"The other side of entropy"*
26. * Condensed Matter Seminar, Dept. of Physics, University of Massachusetts Amherst, May 2024
    Title: *"The other side of entropy"*
27. * Flash talk, CZI 2024 Neuroscience Kickoff Series, Circuitry & Plasticity, Webinar, April 2024
    Title: *"AI Across Biological Scales: From Gene Expression to Behavior"*
28. * Flash talk, NYU AI Research Symposium "Bridging AI Innovation and Societal Impact", New York, April 2024
29. * Minisymposium on "Suppression and Variability in Visual Cortex", Society for Neuroscience annual meeting, Washington D.C., Nov 2023
    Title: *"ORGaNICs: A Recurrent Circuit Theory of Normalization"*
30. * Colloquium, Center for Computational Neuroscience, Flatiron Institute, Nov 2023
    Title: "*Recurrent Circuit Theory of Cortical Communication"*
31. * Machine Learning Potentials - StAtus and FuturE (MLP-SAFE), Virtual Workshop, Jul 2023
    Title: *"ColabFit: An Integrated Platform for Designing, Training, Sharing, and Deploying Machine Learning Interatomic Potentials at Scale."*
32. * Conclave on Complexity in Physical Interacting Systems, Computation, and Thermodynamics, Los Alamos National Laboratory, Santa Fe (NM), Jul 2023
    Title: "*Play. Pause. Rewind. Linking local entropy production, fluxes, and extractable work in active matter."*
33. Energy Landscapes Workshop (Long talk, 1h), Porquerolles (France), Jun 2023
    Title: *"From Slices to Volumes: Unveiling the Geometry of High-Dimensional Energy Landscapes.*"
34. * Complex Systems Seminar, Dept. of Physics, Northwestern University, May 2023
    Title: *"Cortical circuit theory of interareal communication"*
35. * Symposium: Systems for Fitting, Uncertainty Quantification, Selection and Use of Interatomic Models, Mach Conference, April 2023
    Title: *"Colabfit: An integrated Platform for Designing, Training, Sharing, and Deploying Machine Learning Interatomic Potentials at Scale"*
36. * Webinar, Scientific Machine Learning Webinar Series, CMU Arpa-E, March 2023
    Title: *"Colabfit: An integrated Platform for Designing, Training, Sharing, and Deploying Machine Learning Interatomic Potentials at Scale"*

37. * Levich Institute Seminar, CUNY, March 2023
    Title: *"The other side of entropy"*
38. * Invited talk, Focus session: "Information theory and Physics", APS March meeting 2023
    Title: *"The other side of entropy"*
39. * Invited tutorial, Short Course "Computational modeling and information theory: from grains to bits", APS March meeting 2023
    Title: *"The other side of entropy"*
40. * Computations in Science Seminar, Dept. of Physics, University of Chicago, Jan 2023
    Title: *"The other side of entropy"*
41. * 123rd Statistical Mechanics Conference, Rutgers University, Dec 2022
    Title: "*Play. Pause. Rewind. Model Free Measurement of Local Entropy Production and Extractable Work in Active Matter.*"
42. * Colloquium, Center for Computational Mathematics, Flatiron Institute, Dec 2022
    Title: *"The other side of entropy"*
43. * Research Seminar, Santa Fe Institute, Nov 2022
    Title: *"The other side of entropy"*
44. * Research Seminar, Center for Nonlinear Studies, Los Alamos National Laboratory, Nov 2022
    Title: "*Play. Pause. Rewind. Model Free Measurement of Local Entropy Production and Extractable Work in Active Matter.*"
45. * Research Seminar (remote), Dept. of Chemistry, UT Austin, Host: Devarajan Thirumalai, Oct 2022
    Title: *"The other side of entropy"*
46. Frontiers in Computational Chemistry: The NYU Simons Center Inaugural Symposium, Oct 2022
    Title: "*Play. Pause. Rewind. Model Free Measurement of Local Entropy Production and Extractable Work in Active Matter.*"
47. * Theoretical Chemistry Seminar, Dept. of Chemistry, University of Cambridge, Aug 2022
    Title: *"The Other Side of Entropy"*
48. Physics of Information workshop, Dept. of Physics, New York University, May 2022
    Title: *"The Other Side of Entropy"*
49. * Research Seminar (remote), Condensed Matter Theory group, Dept. of Physics, University of Oxford, Host: Shivaji Sondhi, Apr 2022
    Title: *"The other Side of entropy"*
50. APS March meeting, Session D20: Active Matter in Complex Environments II, Chicago, Mar 2022
    Title: *"Play. Pause. Rewind. Measuring local entropy production and extractable work in active matter"*
51. AIChE Annual Meeting, Session: Active Colloidal Systems, Boston, Nov 2021
    Title: *"Play. Pause. Rewind. Measuring local entropy production and extractable work in active matter"*
52. APS March Meeting (remote), Session Y06: Memory Formation in Matter: Encoding, Reading, and Design II, Mar 2021
    Title: *"Squeezing order out of disorder"*
53. * Research Seminar, Dept. of Chemistry, New York University, Mar 2021
    Title: *"The other side of entropy"*

54. * Machine Learning Seminar, University of Minnesota - Twin Cities, Dec 2020
    Title: "*Design and Discovery in the Protein Fitness Landscape*"
55. Applied Physics "Socioloquium" (remote), SEAS, Harvard University, Main speaker: P.M. Chaikin, Contributors: **S. Martiniani**, D. Frenkel, Host: Eric Mazur, Jun 2020
    Title: *"Information and Order"*
56. * Seminar, Chemical Theory Center, Dept. of Chemistry, University of Minnesota - Twin Cities, May 2020
    Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture."*
57. * APS March Meeting, Denver (CO), Session J31: Workshop: Grand Challenges in Soft Matter and Opportunities for Microgravity Research, Mar 2020 (meeting cancelled for COVID-19 crisis)
    Title: *"Squeezing order out of disorder"*
58. * Geomechanics Seminar, Dept. of Civil, Environmental and Geo- Engineering, University of Minnesota - Twin Cities, Feb 2020
    Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
59. * Condensed Matter Seminar, Dept. of Physics, University of Minnesota - Twin Cities, Dec 2019
    Title: *"Squeezing order out of disorder"*
60. * Solid Mechanics Research Seminar, Dept. of Aerospace Engineering and Mechanics, University of Minnesota - Twin Cities, Nov 2019
    Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
61. 17th Annual Northeastern Granular Materials Workshop, New York University, Jun 2019
    Title: *"Taking the numerical calculations of granular entropy forward"*
62. APS March Meeting, Boston, Mar 2019
    Title: *"Characterizing elusive correlation lengths by computable information density"*
63. * Research Seminar, Centre for Theoretical Biological Physics, Rice University, Feb 2019
    Title: *"Quantifying hidden order out of equilibrium"*
64. * Research Seminar, Dept. of Chemical Engineering and Material Science, University of Minnesota - Twin Cities, Feb 2019
    Title: *"Quantifying hidden order out of equilibrium"*
65. * Research Seminar, Dept. of Chemistry, University of Illinois Urbana-Champaign, Jan 2019
    Title: *"Quantifying hidden order out of equilibrium"*
66. * Research Seminar, Santa Fe Institute, Jan 2019
    Title: *"Quantifying hidden order out of equilibrium"*
67. * Research Seminar, Dept. of Chemistry, University of Colorado Boulder, Dec 2018
    Title: *"Quantifying hidden order out of equilibrium"*
68. MRSEC Seminar, New York University, Oct 2018
    Title: *"Quantifying hidden order out of equilibrium"*
69. 16th Annual Northeastern Granular Materials Workshop, Yale University, Jun 2018
    Title: *"Quantifying hidden order out of equilibrium"*
70. * DAMPT Statistical Physics and Soft Matter Seminar, University of Cambridge, May 2018
    Title: *"Quantifying hidden order out of equilibrium"*

71. * Mach Conference, Annapolis (MD), Apr 2018
    Title: *"Taking the numerical calculation of granular entropy forward: a new look at the yielding transition"* - **session keynote**
72. * Army Research Lab (ARL), Aberdeen Proving Ground (MD), Apr 2018
    Title: *"Quantifying hidden order out of equilibrium"*
73. APS March Meeting, Los Angeles (CA), Mar 2018
    Title: *"Quantifying hidden order out of equilibrium"*
74. Edwards Centre for Soft Matter Inaugural Meeting, University of Cambridge, Jan 2017
    Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
75. Recent Advances on the Glass and Jamming Transition, CECAM Workshop, Lausanne, Switzerland, Jan 2017 - **poster prize talk**
    Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
76. * 3rd International Conference on Packing Problems, Shanghai, China, Sept 2016
    Title: *"Turning intractable counting into sampling: Computing the configurational entropy of three-dimensional jammed packings"*
77. StatPhys 26, Lyon, France, Jul 2016
    Title: *"Turning intractable counting into sampling: Computing the configurational entropy of three-dimensional jammed packings"*
78. * Institute for Multiscale Simulations, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany, Aug 2015
    Title: *Turning intractable counting into sampling: the mean basin volume method"*
79. Gordon Conference on Granular and Granular-Fluid Flow, Stonehill College, Easton (MA), Jul 2014
    Title: *"Numerical Calculation of Granular Entropy"*

Posters

1. Lake Conference - Neural Coding and Dynamics, Seattle, Sept 2025
   Title: "*Divisive Normalization and Stability in Recurrent Neural Networks"*
2. Classical and Quantum Complexity in Statistical Mechanics, Molveno (Italy), Jul 2025
   Title: *"Random Organizing Systems: from Physics to AI"*
3. GRC Neurobiology of Cognition: Neural Circuits Supporting Cognitive Function, Newry (ME), Jul 2022
   Title: *"Towards a unifying and analytically tractable framework for neural dynamics"*
4. IEEE European School of Information Theory, Bertinoro, Italy, May 2018
   Title: *"Quantifying hidden order out of equilibrium"*
5. Recent Advances on the Glass and Jamming Transition, CECAM Workshop, Lausanne, Switzerland, Jan 2017
   Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
6. Soft Matter – Theoretical and Industrial Challenges, University of Cambridge, UK, Sept 2016
   Title: *"Are some packings more equal than others? A direct test of the Edwards conjecture"*
7. International Workshop on Jamming and Granular Matter, Queen Mary University of London, UK, Jul 2016

Title: *"Turning intractable counting into sampling: Computing the configurational entropy of three-dimensional jammed packings"*

8. Machine Learning Summer School, Kyoto, Japan, Aug 2015
   Title: *"Computing extremely high-dimensional volumes"*
9. Italian Soft Days, Università la Sapienza, Rome, Italy, Sept 2014
   Title: *"Numerical Calculation of Granular Entropy"*
10. Advanced School in Soft Condensed Matter 'Solutions in the Spring', Homerton College, Cambridge, UK, Apr 2014
    Title: *"Superposition Enhanced Nested Sampling"*
11. International Soft Matter Conference, Università la Sapienza, Rome, Italy, Sept 2013
    Title: *"Superposition Enhanced Nested Sampling"*

Organized workshops and conferences

1. Program Chair, Workshop, AI for Accelerated Materials Design, ICLR, Rio de Janeiro, April 2026
2. Invited Moderator, National Academies' Condensed Matter and Materials Research Committee Workshop: "Frontiers of Materials That Learn", Washington D.C., Oct 2025
3. Member, International Program Committee, 2027 International Soft Matter Conference, San Diego
4. Program Chair, Workshop, AI for Accelerated Materials Design, NeurIPS, San Diego, Dec 2025
5. Organizer, Conference, NYC AI4Chemistry Summit, New York, Jun 2025
6. Program Chair, Workshop, AI for Accelerated Materials Design, ICLR, Singapore, Apr 2025
7. Program Chair, Workshop, AI for Accelerated Materials Design, NeurIPS, Vancouver, Dec 2024
8. Program Chair, Workshop, AI for Accelerated Materials Design, BOKU, Vienna, July 2024
9. Organizer, Working group: "Theory and circuits of cortical function and corticocortical communication", Aspen Center for Physics, Jul 2023.
10. Organizer, Workshop: "Physics of Information", Dept. of Physics, New York University, May 2022
11. Organizer, Focus Session: "Machine Learning for Biomolecular Design and Simulation", APS March Meeting 2021

Extended stays

1. Aspen Center for Physics, "Working group: Theory and circuits of cortical function and corticocortical communication", Jul 2023
2. Aspen Center for Physics, Workshop: "Learning Dynamical Models from Biophysical Data", Jun 2022

Teaching

2022-2025 (NYU)

CHEM-UA 652 "Thermodynamics & Kinetics" (Fall 2022, Fall 2024, Fall 2025)

PHYS-UA 2061 "Mathematical Physics" (Spring 2024)

PHYS-GA 2061 "Physics of Neural Systems" (Spring 2023)

2019-2021 (UMN)	**CHEN 3201** "Numerical Methods for Chemical Engineering" (Spring 2020, co-instructor; Spring 2021, Fall 2021 lead instructor)
	CHEN 8754 "Quantitative Analysis, Design and Synthesis of Biotechnological Systems." Delivered 2.5h of content on "Information theory for biochemical signal pathways" (Spring 2021, contributing lecturer)
	CHEN 5595 "Data Driven Discovery for the Chemical Sciences" (Fall 2020)
	MATS 3001 "Thermodynamics of Materials" (Fall 2019, co-instructor)
	Departmental Workshop (year-long) on Machine Learning for Science ($\sim$ 40 participants).
2013 - 2015	**Mathematics tutor** for natural sciences and computer science students in the Tripos IA (first year) at Magdalene College, Cambridge. Tutorials consisted of year-long 1 hour weekly meetings with 3 or 4 groups of 2 students each.
2013	**Laboratory demonstrator** for Tripos IA (first year) organic chemistry practicals (University of Cambridge).

University Service

NYU FAS	Member, Computational Science Initiative Committee	2023
NYU Chemistry	Member, DEI Committee	2024 - 2025
	Member, Awards Committee	2024 - 2025
	Member, Simons Center Core Team	2022 - 2025
	Member, Colloquium Committee	2022 - 2024
	Member, Theory Faculty Search Committee	2022 - 2023
	Member, Graduate Admission Committee	2023
	Member, Simons Center Administrator Search Committee	2022
NYU Physics	Member, Colloquium Committee	2023 - 2024
	Member, Graduate Admission Committee	2024
	Member, Grant Administrator Search Committee	2024
	Member, CSMR Faculty Search Committee	2022

Other Service

Conference reviewer	AI4Mat ICLR; AI4Mat NeurIPS; NeurIPS	2025
	AI4Mat NeurIPS; AI4Mat Vienna	2024
	COSYNE	2023
Journal reviewer	PRL, PRX, PRE, PNAS, Nat. Neuroscience, J. Chem. Phys., J. Stat. Phys, J. Comp. Phys.	
Editorial roles	Co-founder and member of organizing committee of KIMReview	2023 – present
Study sections	NSF Molecular Foundations for Biotechnology	2022
Other	Gates Cambridge Junior Treasurer, University of Cambridge	2016 – 2017

Synergistic activities

Outreach

1. Organizer of Public Lectures in Foreign Languages, NYU (2025)
2. Public School 3, grade K science hour, New York, NY (2024)
3. Member of Public School 3 STEAM committee, New York, NY (2023 - 2024)
4. NYU Proud to Be First Advocate and Proud to Be First Mentor;
5. Host for high-school students seeking research experience (2022, 2023);
6. Jury member for "Science Court", an interdisciplinary course in the University of Minnesota Honors Program designed to combat polarization in American society and strengthen democracy (2020);
7. Public School 154 STEM Professionals Day, Brooklyn, NY (2019).

Dissemination

1. Video entry at APS-DSOFT Gallery 2024 - awarded jury prize.
2. Video entry at APS-DSOFT Gallery 2022.

Open Science

1. Lead PI of FERMat project, developing Machine Learning "foundation" models for materials and molecular discovery, and associated cyberinfrastructure.
2. co-PI of Colabfit Exchange project, providing the most diverse public database of first principle data for training ML Interatomic Potentials.

Media

1. Research highlighted by scientific and news media worldwide, including in Nature, Nature Materials, Physics (1, 2, 3), Physics Today (1, 2), New Scientist, Science & Vie, Sky News, ANSA.
2. Press-release by U.S. Senate Majority Leader Chuck Schumer and U.S. Senator Kirsten Gillibrand announcing the FERMat project.

Interviews

1. Main guest on the program "Si può fare" by Radio24 (radio station of the leading Italian financial newspaper "Il Sole 24 Ore", 2.2M listeners per day) to talk about Artificial Intelligence and foundation models for materials discovery. The interview is available from minute 20 of this podcast.

Public Awards

"Martin de Segura Prize for Artistic and Cultural Contributions", City of Martinsicuro, Italy (2024)

Current and past group members

PhD

Satyam Anand

Ph.D. Mathematics, 2025, Courant Institute of Mathematical Science
Thesis: "Transport, Energetics, and Self-organization in Non-equilibrium Systems"
Awards: 1) Rising Stars in Soft and Biological Matter, UC San Diego**; 2)** Finalist, APS DSNP graduate student speaker award**; 3)** Martin and Sarah Leibowitz Graduate Prize for Quantitative Biology 2025 (Courant Institute). **4)** Division of Soft Matter Future Investigator Travel Award, American Physical

	Society (2025). **5)** Dean's Conference Travel Award, New York University (2022, 2024). **6)** Travel Award, Gordon Research Conference (GRC): Soft Condensed Matter Physics (2023). **7)** Travel Award, Gordon Research Conference (GRC): Stochastic Physics in Biology (2025).
Praharsh Suryadevara	Ph.D. Physics, 2025 , New York University **Thesis:** "Structure of Basins of Attraction of Soft Sphere Packings" **Awards:** Dean's Conference Travel Award, New York University (2024)
Aaron Shih	Ph.D., Mathematics, 2025, Courant Institute of Mathematical Science **Thesis:** "A Never-Before-Seen World of Correlated Disorder: Functional Metamaterials For Wave Scattering." **Awards: 1)** Gallery of Soft Matter Prize 2024 (APS DSOFT). **2)** Division of Soft Matter Future Investigator Travel Award, American Physical Society (2023). **3)** Dean's Conference Travel Award, New York University (2023).
Shivang Rawat	Ph.D., Mathematics, 2025, Courant Institute of Mathematical Science, **Thesis:** "Divisive Normalization as a Mechanism for Stable and Efficient Recurrent Neural Computation." **Awards: 1)** Wilhelm T. Magnus Memorial Prize 2025 for significant contributions to the mathematical sciences (Courant Institute). **2)** Division of Biological Physics Future Investigator Travel Award, American Physical Society (2024).
Alexander Golinski	Ph.D. Chemical Engineering, 2021, University of Minnesota, co-advised with Ben Hackel (primary). **Thesis:** "Data-Driven Approach to Engineering Protein Evolvability and Developability." **Awards:** NSF GRFP.
Zachary D. Schmitz	Ph.D. Chemical Engineering, 2025, University of Minnesota, co-advised through year 3 with Ben Hackel (primary). **Thesis: "**Data Driven Protein Scaffold Developability Engineering."
Mia Morrell	Physics Ph.D. student, co-advised with David Grier (primary), anticipated graduation 2026.
Asit Pal	Chemistry Ph.D. student, anticipated graduation 2027. **Awards:** Dean's Conference Travel Award, New York University (2024).
Tom Egg	Chemistry Ph.D. student, anticipated graduation 2028. **Awards: 1)** Spotlight Talk, AI4Mat Workshop, International Conference on Machine Learning (ICML), Singapore, 2025; **2)** GAAN Crystal Science Fellowship, NYU; **3)** Predoc Fellowship, Initiative for Computational Catalysis - Simons Foundation
Elijah House	Physics Ph.D. student, anticipated graduation 2028.
Akshada Pradhan	Chemistry Ph.D. student, anticipated graduation 2029.

Jacob Abraham	Physics Ph.D. student, anticipated graduation 2030.
Huijie Chen	Physics Ph.D. student, anticipated graduation 2030.
Shannon Yu	Cognition and Perception Ph.D. student, anticipated graduation 2030, co-advised with J.A. Winawer and D. Heeger (primary).

Postdoctoral and Research Scientists

Philipp Höllmer	Simons Center Fellow, Dept. of Chemistry, NYU, 2024 - present **Awards:** Spotlight Talk, AI4Mat Workshop, International Conference on Machine Learning (ICML), Singapore, 2025
Jiyeon Han	Postdoctoral Associate, Dept. of Physics/CNS, NYU, 2024 - present
Maya Martirossyan	Postdoctoral Associate, Dept. of Physics/Chemistry, NYU, 2024 - present **Awards: 1)** Spotlight Talk, AI4Mat Workshop, International Conference on Machine Learning (ICML), Singapore, 2025. **2)** NYU Arts & Science Postdoctoral Travel Award (2025).
Kathryn Mcclain	Postdoctoral Associate, Dept. of Physics/CNS, NYU, 2025 - present
Rishabh Sharma	Simons Center Fellow, Dept. of Chemistry, NYU, 2025 - present
Tianhao Li	Simons Center Fellow, Dept. of Chemistry, NYU, 2025 - present
Flaviano Morone	Research Scientist, Dept. of Physics/CNS, NYU, 2024 -present
Mathias Casiulis	Research Scientist, Dept. of Physics/Chemistry, NYU, 2022 - 2025. Research Scientist, Dept. of Physics/Chemistry, NYU, 2025 - present. **Awards: 1)** Poster Prize 9th Edwards Symposium, U. Cambridge (2025). **2)** Gallery of Soft Matter Prize 2024 (APS DSOFT).
Eric Fuemmeler	Research Scientist, Dept. of Aerospace Engineering & Mechanics, UMN, 2021 - present (co-advised with Ellad Tadmor) **Awards:** Spotlight Talk, AI4Mat Workshop, NeurIPS, Vancouver, 2024
Guanming Zhang	Postdoctoral Associate, Dept. of Physics/Chemistry, NYU, 2022 - 2025
Amit Gupta	Research Scientist, Dept. of Aerospace Engineering & Mechanics, UMN, 2021 - 2025 (co-advised with Ellad Tadmor)

Other Research Assistants

Gregory Wolf	Software Developer, Dept. of Physics, NYU, 2023 - present
Josh Vita	PhD Research Intern, Dept. of Aerospace Engineering & Mechanics, UMN, 2021-2022 (co-advised with Ryan Elliot and Ellad Tadmor)
Alexander Tao	Research Intern, Dept. of Aerospace Engineering & Mechanics, UMN, Summer 2022 (co-advised with Ellad Tadmor)
Dylan Bruesehoff	Research Intern, Dept. of Aerospace Engineering & Mechanics, UMN, Summer 2022 (co-advised with Ellad Tadmor)

Master

Shenga Zhao	Physics MS student, anticipated graduation 2027.
Zayyam Mohammed	Financial Mathematics MS student, UMN, 2020-2021.

Undergraduate

Zeshan Hulit	Visiting Student, Dept. of Physics, NYU, Summer 2025
Kosta Dubovskiy	Student, Dept. of Physics, NYU, Summer 2025
Kyle Lam	SURP Student, Dept. of Chemistry, NYU, Summer 2024
Stefan Rankovic	Student, Dept. of Physics, NYU, Summer 2024
Asaf Greenfield	Student, Dept. of Physics, NYU, Summer 2024
Kyrie Xie	Student, Courant Institute, NYU, Summer 2022
Carmel Pe'er	AM-SURE Student, Courant Institute, NYU, Summer 2023
Orion Runjia Yang	AM-SURE Student, Courant Institute, NYU, Summer 2023
Charlie Chen	AM-SURE Student, Courant Institute, NYU, Summer 2023 **Notes:** co-authored 1 paper.
Bryce Johnson	Student, Dept. of Physics, UMN 2020-2021. **Notes:** co-authored 1 paper and delivered talk at APS March Meeting 2021.
Diya Saha	Student, Dept. of Chem. Eng. and Materials Science, UMN Summer 2020 **Notes:** co-authored 1 paper.
Sandya Appiah	Student, Dept. of Chem. Eng. and Materials Science, UMN Summer 2020 **Notes:** co-authored 1 paper.
Daniel Ribeiro	Student, Dept. of Chem. Eng. and Materials Science, UMN 2020-2021
Jiwon Kim	Honors Thesis Student, Dept. of Chem. Eng. and Materials Science, UMN 2020-2021
Remi Bougie	Honors Thesis Student, Dept. of Chem. Eng. and Materials Science, UMN 2020-2021
Vijay Vallurupalli	Student, Dept. of Chem. Eng. and Materials Science, UMN Summer 2021
Kovic Odhiambo	Student, Dept. of Statistic, UMN Summer 2020

High school

Tom Zhang	Dept. of Chemistry/Physics, NYU, Summer 2022 **Notes:** co-authored 1 paper. Gone to Caltech (Computer Science).
Juhee Park	Dept. of Chemistry/Physics, NYU, Summer 2022 **Notes:** Gone to Carnegie Mellon University (Chemical Engineering).
Gabriel Magnasco-Farinas	Dept. of Chemistry/Physics, NYU, Summer 2023 **Notes:** Gone to University of Colorado Boulder (Mathematics and Economics).

Milan Lustig	Dept. of Chemistry/Physics, NYU, Summer 2024 **Notes:** Regeneron Science Talent Search Scholar Semifinalist and Regeneron STS Top 300 Scholar (2025). Gone to Harvard (Computer Science and Electrical Engineering).

2014-2019

U. Cambridge	**Day-to-day supervisor** for 2 Physics Part III final year project (one awarded Part III prize for best computational physics project); 1 exchange master student (awarded distinction from TU-Munich).

Other professional experience

Cadet (active duty)	Italian Army, "Scuola Militare Teulié", Milan	2005 – 2008